

Business Plan Proposal

Drone-Enabled Illegal Dumping Detection & Rapid Waste Removal Service

Proponent: Greenback Recovery PTY LTD

Client: City of Wanneroo

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1. Executive Summary

This proposal outlines a drone-enabled, end-to-end illegal dumping detection and rapid waste removal service that supports the City of Wanneroo's priorities for liveable communities, environmental stewardship, regulatory compliance, and community trust.

Greenback Recovery Pty Ltd will act as the **prime service provider**, delivering detection, documentation, mobilisation, removal verification, and council-ready reporting. Physical waste removal, transport and disposal will be completed by **Greenback's prequalified subcontractor network**, operating under Greenback's safety and compliance systems. All works will be supported by an evidence pack including **GPS and time-stamped photos, weighbridge dockets, and disposal receipts**.

Greenback has already identified and mapped illegal dumping locations within the **Gnangara forest area**, with an estimated total volume of approximately **100 tonnes of mixed waste**. The immediate priority is to clear this existing waste efficiently and restore affected areas. Following the initial remediation, Greenback will use drones to conduct ongoing scanning of nominated hotspots to identify further dumping quickly and reduce reliance on public reporting, saving time and council resources.

Phase 1 (Detection & Removal – Core Service): Drone scanning, GPS mapping, waste verification and removal delivery for identified sites.

Phase 2 (Reporting & Documentation – Included): Site-by-site compliance documentation and performance reporting.

Phase 3 (Optional – AI Deterrence System, Separate Quote): Solar-powered AI cameras installed at hotspots to detect vehicles entering dumping zones and notify ranger/compliance teams to reduce repeat dumping.

Pricing approach (indicative and negotiable):

- Initial remediation of the currently mapped ~100 tonnes: **lump sum \$1.18M to \$1.44M (ex-GST), negotiable** subject to joint verification and access conditions.
- After the initial remediation: Free drone scanning of nominated hotspots, with removal charged at **\$8,776 to \$9,661 per tonne (ex-GST)** for any newly detected waste.

Greenback is partnered with **Licence 4 Work** to support workforce training and competency requirements where needed. Greenback's service is designed to remove **A–Z waste types** commonly found in bushland dumping (e-

waste, whitegoods, household goods, metals, mattresses, and mixed debris), while reducing environmental risk, improving response times, and delivering transparent outcomes the city can report to the community.

2. Councilor Brief

Purpose

To provide the City of Wanneroo with a proactive, technology-enabled solution to illegal dumping that protects the environment, strengthens compliance, reduces operational burden, and improves community confidence.

Strategic Alignment

This proposal aligns with core council priorities:

- **Liveable Communities:** Faster clean-up of illegal dumping improves amenity, safety and public access to shared spaces.
- **Environmental Stewardship:** Early detection and removal protects bushland, waterways and native ecosystems.
- **Good Governance and Compliance:** Reduced risk of environmental breaches and improved audit readiness through evidence-based reporting.
- **Community Trust:** Visible action and measurable outcomes reduce complaints and reinforce public confidence.

The Problem

Based on standard local government staffing costs and workflows:

- Average staff time per incident: 2–5 hours
- Estimated annual incidents (metro councils): 1,000–3,000+

This equates to an annual administrative cost of approximately \$150,000 to \$630,000, excluding clean-up expenses. When combined with remediation and removal costs, illegal dumping expenditure can reach hundreds of thousands to several million dollars annually, as demonstrated by multiple Australian councils. Critically, these demands divert council staff from strategic priorities such as environmental planning, compliance enforcement and community engagement.

Immediate Local Context (Wanneroo)

Greenback has already identified and mapped illegal dumping locations within the Gngangara forest area, with an estimated combined volume of approximately **100 tonnes** of mixed waste. This waste creates ongoing environmental, safety and reputational risk while remaining in the landscape.

The Solution

Greenback Recovery will provide:

- **Drone surveillance** of nominated high-risk dumping locations
- **GPS mapping and visual verification** of dumped waste to support prioritisation
- **Centralised mobilisation and management of Greenback's prequalified subcontractor network** for removal, transport and disposal
- **Council-ready reporting and performance metrics**, including evidence packs supported by GPS/time-stamped photos, weighbridge dockets and disposal receipts
- **Optional Phase 3 capability (separate quote):** solar-powered AI camera deterrence at hotspots to identify vehicles entering dumping zones and notify ranger/compliance teams

Key Benefits to Council

- Reduced environmental and public health risk
- Faster response times and improved audit readiness
- Reduced administrative workload for council teams
- Fewer community complaints and improved public confidence
- Better data to guide prevention planning and resource allocation

Recommended Next Step

Approve an initial remediation program to clear the currently mapped ~100 tonnes in Gnangara Forest, followed by ongoing drone scanning of nominated hotspots to identify new dumping early. Post-remediation removal works can be delivered under an agreed per-tonne rate, with clear service measures and quarterly reporting.

3. The Challenge

Illegal dumping continues to increase across metropolitan and peri-urban council areas, particularly in bushland reserves, road verges and semi-isolated access points. The key challenges include:

Environmental and Public Health Risks

- Asbestos exposure from illegally dumped construction materials
- Soil contamination and long-term land degradation
- Leaching of hazardous materials into waterways and groundwater
- Fire risk from flammable waste, particularly in summer
- Harm to wildlife and native vegetation
- Spread of weeds and invasive species

Operational and Governance Challenges

- Heavy reliance on public reporting, which is inconsistent and delayed
- Limited visibility of remote or inaccessible dumping locations
- Administrative overload managing reports, contractors and follow-ups
- Risk of environmental compliance breaches if waste is not removed promptly
- Increasing community complaints and reputational impact for councils

Photo Evidence:

Due to file size limits, only sample photos are included. Full photos and mapping evidence are available via Drive on request.







4. The Solution

Greenback Recovery Pty Ltd proposes a proactive, technology-enabled service that integrates **drone surveillance, GPS mapping, subcontractor-managed removal, and council-ready reporting** into a single accountable delivery model.

Key Components

- **Drone surveillance** of nominated high-risk dumping zones and bushland areas
- **GPS-accurate location mapping** of dumped materials, with site ID allocation
- **Visual verification and classification** to assess waste type and potential risk (e.g., construction waste, e-waste, whitegoods, mattresses and mixed debris)
- **Mobilisation and management of Greenback's prequalified subcontractor network** for waste removal, transport and lawful disposal/recycling
- **Evidence pack for every site**, including GPS/time-stamped before/after photos, weighbridge dockets and disposal receipts
- **Council-ready reporting and compliance documentation**, including summary metrics and quarterly reporting (as agreed)
- **Optional Phase 3 (Separate Quote)**: If Council requires stronger deterrence to reduce future dumping, Greenback can deploy **solar-powered AI camera monitoring** at hotspot entry points to detect vehicles entering dumping zones and notify ranger/compliance teams.

5. Outcomes and Benefits

By outsourcing illegal dumping detection and response to a specialist service, councils achieve:

- Faster response times for waste removal
- Reduced environmental and public health risk
- Lower reliance on public reporting systems
- Reduced Administrative Workload for council teams
- Improved compliance outcomes and audit readiness

- Decreased community complaints
- Data-driven decision making through repeat-location tracking and analytics

6. The Three-Phase Service Model

Phase 1: Detection & Removal Coordination

- Drone surveillance of identified high-risk zones
- Verification and classification of dumped waste
- Immediate coordination with approved contractors
- Tracking of removal timeframes and completion

Phase 2: Reporting & Documentation (Included with Phase 1)

- Site-by-site evidence packs (GPS/time-stamped photos, location mapping, disposal receipts, weighbridge dockets)
- Tonnes removed and waste stream summary (mixed waste, e-waste, whitegoods, metals, mattresses, etc.)
- Turnaround time from detection to removal
- Hotspot dashboard (locations detected, cleared, and repeat areas)
- Quarterly summary report and annual performance statement (as agreed)

Phase 3: Optional AI Deterrence System (Separate Quote)

If Council requires a stronger long-term deterrence solution to reduce repeat dumping, Greenback can deploy an AI-enabled monitoring system at identified hotspots:

- Solar-powered AI cameras installed at hotspot entry points / access tracks
 - Vehicle detection and alert notifications to ranger/compliance teams when suspicious activity occurs
 - Hotspot trend analytics to support targeted enforcement and prevention planning
- Note: Phase 3 is optional and will be delivered under a separate scope and quotation.*

7. Financial Model (Indicative)

Pricing Structure Overview – to be refined with Council

The service is designed to provide flexibility, scalability and predictable budgeting for council.

Core Service (Phase 1)

- Monthly retainer: \$5,000 – \$30,000 depending on coverage
- OR per-incident fee: \$150 – \$300 per case

Optional Add-On (Phase 2: AI Monitoring)

- Installation: \$3,000 – \$10,000 per hotspot
- Monitoring: \$1,000 – \$3,000 per hotspot/month

Reporting (Phase 3)

- Included in core service (basic)
- Advanced reporting: \$2,000 – \$5,000 per quarter

Waste Removal Costs

- Managed via contractors (pass-through cost)
- Estimated: \$2,500 – \$3,500 per tonne

Pilot Program Option

A 3–6 month pilot program is recommended:

£ \$15,000 – \$60,000 total (depending on scope)

Includes: Targeted hotspot monitoring | Removal coordination | Performance reporting

Note: A detailed cost breakdown and tailored pricing model can be provided based on council land size, risk profile and identified dumping hotspots.

8. Why Outsource vs Internal Management

Local governments are increasingly required to manage complex environmental risks while operating under constrained budgets, staffing limitations and growing community expectations. Managing illegal dumping entirely in-house presents several structural challenges that outsourcing can directly address.

Limitations of Internal Management

- **Reactive, complaint-driven systems:** Internal models rely heavily on public reporting, resulting in delayed detection and inconsistent coverage, particularly in remote or bushland areas.
- **High administrative burden:** Council staff must manage reports, verify locations, engage contractors, track removals, and respond to follow-up complaints—diverting resources from core strategic work.
- **Specialist capability gaps:** Drone operations, AI-assisted analysis, and repeat-location tracking require specialist licensing, technology, training and ongoing compliance.
- **Increased risk exposure:** Delays in identifying and removing hazardous waste (e.g. asbestos) increase environmental, health and legal risk for councils.
- **Cost inefficiency at scale:** Maintaining internal drone fleets, software systems and trained operators is capital-intensive and difficult to justify for intermittent use.

Advantages of Outsourcing This Service

- **Immediate access to specialist expertise** without capital investment in equipment, licensing or training
- **Proactive detection model** that identifies dumping before complaints escalate
- **Single point of accountability** for detection, coordination, removal tracking and reporting
- **Faster response times**, reducing environmental harm and compliance risk
- **Predictable costs** through retainers or service-based pricing
- **Scalable and flexible delivery**, allowing councils to expand or reduce coverage based on seasonal risk or budget
- **Data-led insights** that internal systems typically do not capture, including hotspot analysis and trend forecasting

Strategic Value to the City of Wanneroo

Outsourcing illegal dumping management directly supports the City of Wanneroo's strategic priorities of environmental protection, regulatory compliance and community trust.

Environmental Protection Local examples in Perth's northern corridor (including areas near Wanneroo) have shown dumping occurring in state forests and bushland reserves, requiring enforcement and clean-up responses.

A proactive detection and rapid response model significantly reduce the time waste remains in the environment, directly supporting the City's commitment to environmental stewardship and sustainable land management.

Compliance and Risk Management The DWER Waste Crime program reported **159 illegal dumping investigations in 2024–25** with multiple prosecutions and infringements and average penalties of **~\$18,825**, with some cases reaching **\$115,000** ([Waste Authority Annual Report 2024-2025](#))

By improving detection speed and maintaining accurate, auditable records (GPS tracking, timestamps, contractor verification), Community Trust and Satisfaction Illegal dumping is a highly visible issue that directly affects residents' perception of council performance. Proactive monitoring and faster clean-ups reduce community complaints, improve amenity and reinforce public confidence that the city is protecting shared spaces. Transparent reporting also enables the city to communicate outcomes and improvements back to the community.

Efficient Use of Public Resources Illegal dumping (verge and built up areas) imposes a significant financial burden on WA councils with estimated cost to local governments of up to \$1 million annually ([WAMNNNews, 2017](#)) on illegal dumping clean-up and management.

9. Recommendations

We recommend the City of Wanneroo:

- **Approve Phase 1 (Detection & Removal)** to clear the already identified and mapped illegal dumping sites in the Gnangara forest area (approximately **100 tonnes**) under an agreed lump sum pricing arrangement (negotiable subject to joint verification).
- **Adopt Phase 2 (Reporting & Documentation)** as a standard requirement for all removals, including evidence packs supported by GPS/time-stamped photos, weighbridge dockets and disposal receipts, with quarterly performance reporting.

- **Endorse ongoing drone scanning** of nominated hotspots after the initial remediation to enable early detection of new dumping events and reduce reliance on public reporting (provided at no cost as part of the post-remediation model).
- **Agree to a per-tonne rate** for any newly detected waste removed after the initial remediation, supported by documented weights and disposal evidence.
- **Consider Phase 3 (Optional AI Deterrence System)** as a separate future scope if Council seeks a stronger prevention solution, including solar-powered AI cameras at hotspot entry points to notify ranger/compliance teams.
- **Evaluate scalability** across neighbouring councils to reduce displacement effects and improve regional dumping control

Illegal dumping is a complex, growing challenge that requires modern, proactive solutions. This proposal offers a practical, scalable and environmentally responsible service that reduces risk, improves response times and delivers measurable results for councils and communities.

We welcome the opportunity to further discuss a implementation with the City of Wanneroo and explore broader regional adoption.

Prepared by: Greenback Recovery PTY LTD